

Complement clauses in Turkish

How much *saying* is necessary?

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IMMAGES workshop
Eberhard Karls Universität Tübingen
Oct. 11–12, 2024

Acknowledgments

The work from today draws on Özyıldız & Uegaki (2024) and on ongoing work with Wataru Uegaki, Faruk Akkuş & Travis Major.

It is made better by audiences in Toronto, New Orleans, Konstanz, Edinburgh and here, and by conversations with Tanya Bondarenko, David Diem, Richard Faure, Miray Gökkaya, Jérémy Pasquereau, Maribel Romero & Marc-Matthias Zymla.

Turkish has a wide range of embedded clauses

1. Nominalizations

- (1) [Biz-im var-dıĝ-ımız-ı] düşün-üyor.
1P-GEN arrive-NMZ-1P.POSS-ACC think-PRES.3S
She thinks that we arrived.

Nominalized clauses feature:

- ✧ one of three nominalizing morphemes, 'indicative' {non-future, future} or a 'subjunctive'
- ✧ genitive subjects, usually + possessive agreement,
- ✧ case.

These can translate *that*-clauses and infinitives.
That is, (1) $\neq$ 'think about your arrival.'

Turkish has a wide range of embedded clauses

2. *Diye* clauses

(2) [Biz var-dı-k diye] düşün-üyor.
1 P.NOM arrive-PST-1 P DIYE think-PRES.3S

She thinks that we arrived.

cf. Biz vardık (*diye).
We arrived.

- ✧ *Diye*-clauses look like root clauses.
They can but need not be direct quotes, e.g., “*ah!*” *diye*.
- ✧ Morphologically, *diye* is the converbial form of *de-*, ‘say’.
Converbial forms range from productive to opaque, e.g., *göre*.

Turkish has a wide range of embedded clauses

3. Bare clauses

- (3) [Biz var-dı-k] san-ıyor.
1P.NOM arrive-PST-1P believe-PRES.3S
She believes that we arrived.

These are ill-understood and we will set them aside.
(Some data in appendix.)

So what?

We want to know more about how clauses compose with the material around them.

Often, the same Turkish predicate combines with 2+ clause types, which gives rise to syntactic and semantic contrasts.

By manipulating things, we can determine what properties to attribute to clauses, or to the predicates they compose with, or to their modes of composition.

Effects of clause type choice

Keeping lexical material constant, NMZ + *V* and *diye* + *V* sometimes share readings, like (1) and (2).

Often, clause type choice makes clear interpretive differences.

Indexical shift

Possible in *diye* clauses, impossible in nominalizations.

Factivity

The truth of a *diye* clause is never* entailed.

With nominalizations, this depends on the verb.

Linguistic production

Diye clauses sometimes give rise to the inference that their content was uttered, thought, entertained, etc.

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Illustrating linguistic production

- (4) [Kim geldi diye] zıpladı.
who arrived DIYE jumped up
She jumped, while saying/wondering: “Who came?”

Illustrating linguistic production

- (5) [Hile yap-tığ-ın-ı] inkâr etti.
cheat do-NMZ-3S-ACC deny
He denied that he cheated. (↪ He said he didn't cheat.)

- (6) (Birşeyi) [hile yaptı diye] inkâr etti.
something.ACC cheat did DIYE denied
He denied something, in saying that he cheated.

Illustrating linguistic production

- (7) [Hile yaptı diye] biliyor.
cheat did DİYE thinks
He thinks that he cheated.

no linguistic production

Two guiding hypotheses

partially independent from one another

✧ Syntax

Nominalizations are **arguments**.

Diye clauses are **adjuncts**.

(to, e.g., vP)

✧ Semantics

Diye clauses mean “**to say, and**”.

Nominalizations get their meaning from their functor.

$$[YP\ X]_{XP} \rightsquigarrow \llbracket X \rrbracket(\llbracket YP \rrbracket)$$

These hypotheses go a long way.

But... are *diye* clauses always adjuncts, or only sometimes?

Do they always involve linguistic production, or only sometimes?

Some authors have argued that adjunction + Predicate Modification are the only way to compose with clauses (e.g., Elliott 2017).

Moreover, these proposals are formulated for *that*-clauses, which nominalizations are a counterpart of.

The view from Turkish is that we need at least two ways of composing clauses with predicates, whatever the exact implementation may be. (See also Bochnak & Hanink 2022 and Deal 2024 on two classes of clauses in Washo and Nez Perce.)

Argument- vs. adjuncthood

The ‘case on the causee’ test (Çetinoğlu & Butt 2008; Yıldırım-Gündoğdu 2017, 2018; Akkuş 2021)

(8) Core contrast

- a. Travis Jack’i giy-dir-di.
Travis Jack.ACC wear-CAUS-PST.3S
Travis dressed Jack (lit. made him wear).
- b. Travis Jack’e giysi(-yi) giy-dir-di.
Travis Jack.DAT clothes-ACC wear-CAUS-PST.3S
Travis made Jack wear clothes. ... *Jack’i...

Argument- vs. adjuncthood

The 'case on the causee' test

(9) Spot checks

- a. Travis Jack'ı hızlıca giydirdi.
Travis Jack.ACC quickly wear-CAUS-PST.3S
Travis dressed Jack quickly.
- b. Travis Jack'e Ø giydirdi.
Travis Jack.DAT wear-CAUS-PST.3S
Travis made Jack wear it.

Further question: Are there accusative NPs in Turkish that are bona fide adjuncts?

Argument- vs. adjuncthood

The ‘case on the causee’ test applied to nominalizations

Accusative nominalizations obligatorily trigger dative causees.

- (10) Bana [var-dıĝ-ın-ı] düşün-dür-dü.
1S.DAT arrive-NMZ-2S-ACC think-CAUS-PST.3S
She made me think that you arrived.

- (11) *Beni [var-dıĝ-ın-ı] düşün-dür-dü.
1S.ACC arrive-NMZ-2S-ACC think-CAUS-PST.3S
Int. She made me think that you arrived.

⇒ Nominalizations pattern like other arguments in the language, which should ideally be reflected in the semantic composition.

Argument- vs. adjuncthood

The ‘case on the causee’ test applied to *diye* clauses

Diye clauses are compatible with accusative causees.

- (12) Beni [var-dın diye] düşün-dür-dü.
1S.ACC arrive-PST.2S DIYE think-CAUS-PST.3S
She made me think that you arrived.

⇒ This suggests the **availability of an adjunct parse**.

Argument- vs. adjuncthood

The ‘case on the causee’ test applied to *diye* clauses

They are also compatible with dative causees.

(The exact conditions of when/who for this is possible are unclear.)

- (13) Bana [var-dın diye] düşün-dür-dü.
 1S.ACC arrive-PST.2S DIYE think-CAUS-PST.3S
 She made me think that you arrived.

⇒ This suggests the **availability of an argument parse**.

This same test can be run in French as well, and other languages with the relevant ACC/DAT alternation.

- (14) Je lui ai fait {dire, penser} que j'étais arrivé.
I 3S.DAT have made say think that I=was arrived
I made him {say, think} that I had arrived.
- (15) *Je l'ai fait {dire, penser} que...
I 3S.ACC=have made say think that

More evidence for the adjunct parse

It's not enough to invoke the case resistance principle (Stowell 1981)

Diye clauses are free to combine with intransitive predicates.

- (16) [Kim geldi diye] zıpladı.
who arrived DIYE jumped up
She jumped saying/wondering: "Who came?"

Methodological note:

This *diye* clause could be a different from the ones with 'think.'
But, if an adjunct parse is available anyway, give it the broadest empirical coverage possible, e.g., for (12).

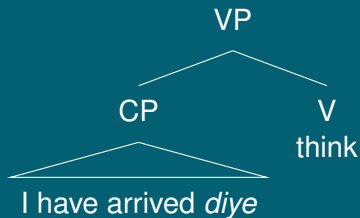
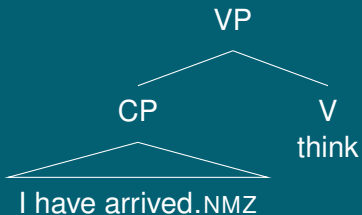
The simplest interpretation of the ‘case on the causee’ tests for argument vs. adjuncthood is:

- ✧ Nominalized clauses are necessarily arguments.
- ✧ *Diye* clauses are sometimes adjuncts and sometimes arguments.

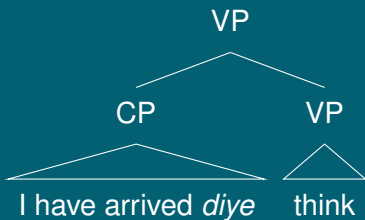
In the remainder of this talk, we will see that. . .

- ✧ allowing for an argument parse for *diye* clauses gives rise to puzzles,
- ✧ but maintaining the view that *diye* clauses are uniformly adjuncts is not fully satisfactory either.

The two argument parses



The adjunct parse



Two puzzles for the argument parse:

#1 *Diye* clauses can't be subjects.

#2 We might expect argument *diye* to have the same denotation as nominalized clauses—this expectation is not generally borne out.

Puzzle #1: *Diye* clauses in subject position?

Diye clauses can occur with passivized predicates.

- (17) [Var-dın diye] düşün-ül-dü.
arrive-PST.2S DIYE think-PASS-PST.3S
It was thought that you arrived.

- ✧ The adjunct parse should be available. Fine.
- ✧ Under the argument parse, the clause in (17) is now a subject.
Then, *diye* clauses should be able to be subjects *in general*.

Discuss Hartman (2012): “DPs, but not CPs can be promoted to subject.”

Puzzle #1: *Diye* clauses in subject position?

But these clauses can't be subjects in general.

- (18) **[Var-dın diye] {doğru, belli, kesin}.*
arrive-PST.2S *DIYE* true obvious certain.COP.3S
Int. It is {true, obvious, certain} that you arrived.

Then they must not be the subject of passive 'think.'

and they must not be the object of active & causative 'think.'

This is also a problem for the adjunct parse (discussion).

Puzzle #1: *Diye* clauses in subject position?

Then what X is such that X gets accusative in (19), for that to trigger dative on the causee?

- (19) Bana [var-dın diye] düşün-dür-dü.
1S.ACC arrive-PST.2S DIYE think-CAUS-PST.3S
She made me think that you arrived.

The contrast between unacceptable subject *diye* clauses and (20) suggests that these can't *freely* be introduced by silent (pro)nouns.

- (20) [[Var-dın diye] bir bilgi] doğru.
arrive-PST.2S DIYE one information true.COP.3S
A piece of information that says that you arrived is true.

Puzzle #2: Interpretive differences with nominalizations

We have to make two assumptions:

- ✧ For the same lexical material L $\llbracket L \text{ } diye \rrbracket = \llbracket L \text{ NMZ} \rrbracket$

“Sometimes, *diye* and nominalized clauses denote the same thing.”

- ✧ $[YP \ X]_{XP} \rightsquigarrow \llbracket X \rrbracket(\llbracket YP \rrbracket)$

Head-complement relationships translate into function-argument relationships.

Puzzle #2: Interpretive differences with nominalizations

- (21) [Gel-diğ-in-i] inkâr et-ti.
come-NMZ-2S-ACC denyPST.3S
She denied that you came.
- (22) [Gel-din diye] inkâr et-ti.
come-PST.2S DIYE denyPST.3S
She denied something, saying that you came.

This difference is hard to capture if we assume that *diye* is the argument of “deny.”

Puzzle #2: Interpretive differences with nominalizations

Question vs. answer orientation

- (23) [Ne ol-duğ-un-u] cevapla-dı.
what be-NMZ-3S-ACC answer-PST.3S
She answered the question of what happened.
- (24) [Ne ol-du diye] cevapla-dı.
what be-PST.3S DIYE answer-PST.3S
She answered a question, saying “What happened?”

A slide for Furkan

Novel data?

It appears that *diye* clauses might be able to be case marked overtly...

- (25) [Ne ol-du diye-yi] cevapla-dı.
what be-PST.3S DIYE-ACC answer-PST.3S
She answered the question of what happened?

To the extent that this is possible, they become synonymous with their nominalized counterparts.

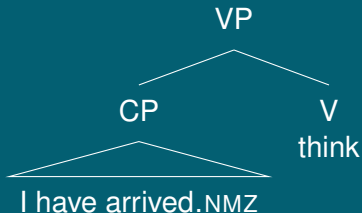
The Özyıldız & Uegaki solution

Assuming the availability of an argument parse makes unwanted predictions.

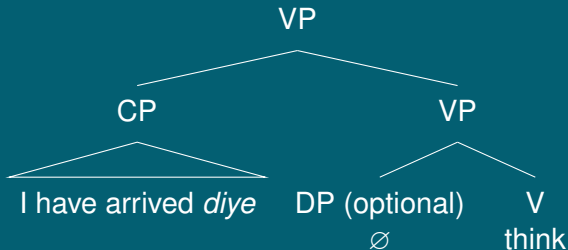
What if we could maintain a uniformly adjunct parse for *diye* clauses?

- ✧ Syntax,
- ✧ Semantics.

The single argument parse



The single adjunct parse (optionality of DP explains ACC/DAT)



We assume that *diye* clauses always contribute a saying event.

$$(26) \quad \llbracket S \text{ diye} \rrbracket = \\ \lambda V_{vt}. \lambda e. \exists e_1, e_2 : \text{say}(e_1, S) \wedge R(e_1, e_2) \wedge V(e_2)$$

- ✧ We take *say* to describe events of (somehow) producing *S*.
- ✧ *R* requires identifying e_1 and e_2 when possible, and summing them otherwise.
- ✧ *V* is saturated by the main event predicate ('think,' 'deny,' 'jump up').

The ‘think’ and ‘deny’ cases

(27) $\llbracket \text{She [I arrived } diye] \text{ deny} \rrbracket =$
 $\lambda e. \exists e_1, e_2, x : say(e_1, \text{I arrived}) \wedge e_1 = e_2 \wedge deny(e_2, x)$

(28) $\llbracket \text{She [I arrive } diye] \text{ think} \rrbracket =$
 $\lambda e. \exists e_1, e_2, x : say(e_1, \text{I arrived}) \wedge e_1 = e_2 \wedge think(e_2, x)$

“She denied/thought (something) and said that I arrived.”

The ‘jump up’ case

$$(29) \quad \llbracket \text{She [I arrived } diye] \text{ jump up} \rrbracket = \\ \lambda e. \exists e_1, e_2 : say(e_1, \text{I arrived}) \wedge e_1 \oplus e_2 \wedge jump\ up(e_2)$$

“She jumped up and said that I arrived.”

In short

We can have an analysis for *diye* clauses that is uniformly adjunctive. . .

whose semantics is weak (summation OR identification),
and relies on null accusative pronouns.

Loose ends:

Left to the audience to uncover.

Conclusions

There are languages with complement clauses that behave differently from each other, syntax and semantics-wise.

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There is much to sort through. But what we can be sure of is:

We need to admit multiple paths to composition.

And/or sort out what's relevant to a given enterprise and what's not.

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There is much to sort through. But what we can be sure of is:

- We need to admit multiple paths to composition.

- And/or sort out what's relevant to a given enterprise and what's not.

- We need to think about the possibility that some classes of clauses have differently patterning subclasses? (Adjunct vs. argument-like *diye* clauses.)

- And how to model them given tools that model synchronic data formulated on the basis of other languages.

Appendix

Bare clauses

Non-interrogative bare clauses are only acceptable with a handful of verbs.

- (30) [Biz var-dı-k] {sanıyor, dedi, *düşünüyor}.
 we arrive-PST-1P believes said thinks
 She {believes, said, *thinks} that we arrived.
- (31) [Biz var-alım] {istiyor, *rica etti}.
 we arrive-OPT.1P wants requested
 She {wants, *requested} that we arrive.

These verbs are also sometimes unhappy with *diye* clauses.

Bare clauses

Interrogative bare clauses enjoy a freer distribution, being acceptable with the set of verbs that generally combine with questions.

- (32) [Biz var-dı-k mı] {düşünüyor, biliyor}.
 we arrive-PST-1P Q thinks knows
 She {is thinking, knows} whether we arrived.

They also coöccur with pronouns.

- (33) [Biz var-dı-k mı] onu {düşünüyor, biliyor}.
 we arrive-PST-1P Q 3S.ACC thinks knows
 She {is thinking, knows} whether we arrived.

Naturally occurring ACC and DAT causees

- (34) Cesedini parçalayıp lögara atmak lazım mezarı bile olmasın **diye düşündürdü beni**

That made me (ACC) think that they shouldn't even have a grave.

- (35) Belki de psikoloji okumasının karakter çözümlemesinde çok büyük faydası var **diye düşündürdü bana.**

That made me (DAT) think that their having studied psychology has a great effect on their character analyses.

Puzzle #1 persists with passivized causatives

Adjunct or ‘case-resisting’ parse

- (36) Sen [vardım diye] düşün-dür-ül-dü-n.
2S.NOM I arrived DIYE think-CAUS-PASS-PST-2S
You were made to think that I had arrived.

Accusative-taking parse

- (37) Sana [vardım diye] düşün-dür-ül-dü.
2S.DAT I arrived DIYE think-CAUS-PASS-PST.3S
You were made to think that I had arrived.